A

Project Report

Submitted in partial fulfilment of the requirements for the completion of the Course

EDS Lab Codetantra

Practical No.5

(2304102L)

Submitted by,

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Under the Guidance of,

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5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

Create a stacked area plot using the data.

Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.

Display the plot showing the temperature variation for each city throughout the months of the year.

```
import matplotlib.pyplot as plt

import pandas as pd


# Data for Months and Temperature for three cities

data = {

    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August',
'September', 'October', 'November', 'December'],

    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],

    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],

    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
```

```
}

# Write your code
```

```
df = pd.DataFrame(data)

plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_B_Temperature'],df['City_C_Temperature'])

plt.xlabel('Month')

plt.ylabel('Temperature')

plt.title('Temperature Variation')

plt.show()
```

5.2.1. Titanic Dataset

16:35

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named titanic.csv and has been loaded using the pandas library. It contains the following columns:

Pclass: Passenger class (1 = First, 2 = Second, 3 = Third).

Gender: Gender of the passenger (male/female).

Age: Age of the passenger.

Survived: Survival status (0 = Did not survive, 1 = Survived).

Fare: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution) :

Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).

Use the color skyblue for the bars.

Title the plot as **"Passenger Class Distribution"** .

Label the x-axis as **"Pclass"** and the y-axis as **"Count"** .

Pie Chart (Gender Distribution) :

Create a pie chart to display the distribution of male and female passengers.

Use lightblue for males and lightcoral for females.

Include percentages on the slices (use autopct='%1.1f%%').

Title the plot as **"Gender Distribution"** .

Histogram (Age Distribution) :

Create a histogram to visualize the distribution of passengers' ages.

Use lightgreen for the bars with black edges (edgecolor = 'black').

Set the number of bins to **8** for the histogram.

Title the plot as **"Age Distribution"** .

Label the x-axis as **"Age"** and the y-axis as **"Frequency"** .

Bar Plot (Survival Count) :

Create a bar plot to show the count of passengers who survived and those who did not, based on the Survived column.

Use the colors lightblue for survivors (1) and lightcoral for non-survivors

(0). Title the plot as **"Survival Count"** .

Label the x-axis as **"Survived (0 =No, 1 =Yes)"** and the y-axis as **"Count"** .

Scatter Plot (Fare vs Age) :

Create a scatter plot to visualize the relationship between the Fare and Age of passengers.

Use orange for the data points.

Title the plot as **"Fare vs Age"** .

Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

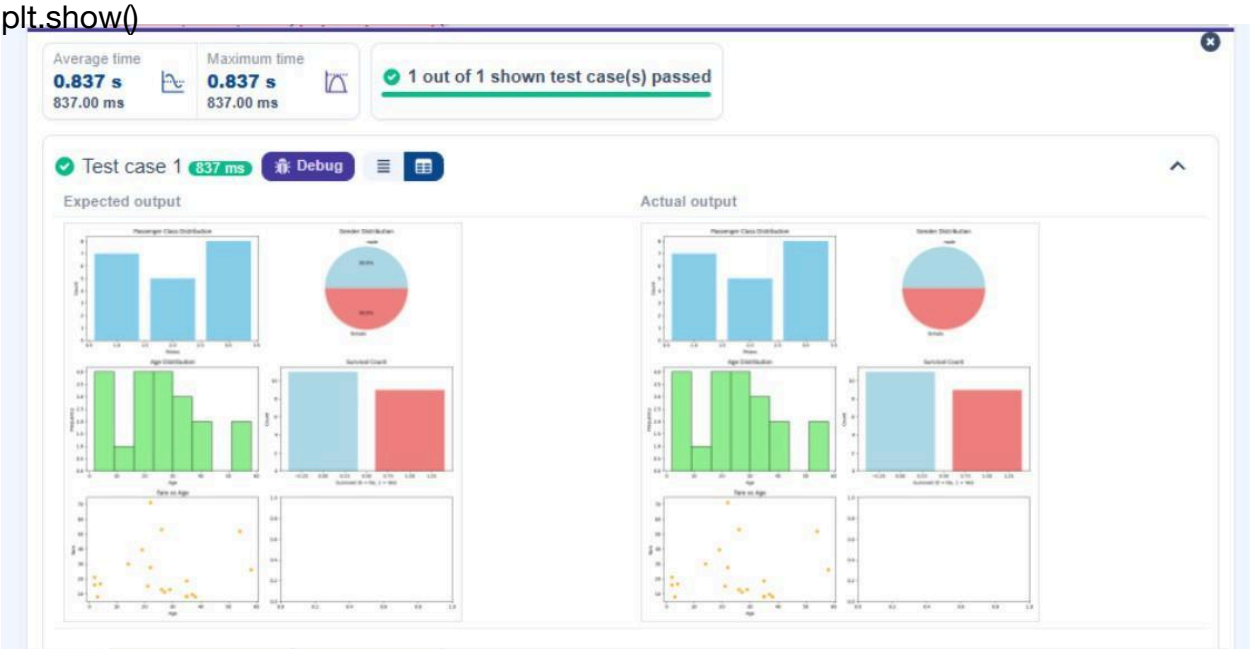
Note: Refer to the displayed plot in the sample test cases for better

understanding. import pandas as pd

```
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))
# write the code..
axes[0,0].bar(df['Pclass'].value_counts().index,df['Pclass'].value_counts(),color='skyblu
e ')
axes[0,0].set_title('Passenger Class
Distribution') axes[0,0].set_xlabel('Pclass')
axes[0,0].set_ylabel('Count')
gender_counts=df['Gender'].value_counts()
axes[0,1].pie(gender_counts,labels=gender_counts.index,colors=['lightblue','lightcoral'])
axes[0,1].set_title('Gender Distribution')
axes[1,0].hist(df['Age'].dropna(),bins=8,color='lightgreen',edgecolor='black')
axes[1,0].set_title('Age Distribution')
axes[1,0].set_xlabel('Age')
axes[1,0].set_ylabel('Frequency')
survival_counts=df['Survived'].value_counts()
axes[1,1].bar(survival_counts.index,survival_counts,color=['lightblue','lightcoral'])
axes[1,1].set_title('Survival Count')
axes[1,1].set_xlabel('Survived (0 = No, 1 = Yes)')
axes[1,1].set_ylabel('Count')
axes[2,0].scatter(df['Age'],df['Fare'],color='orange')
axes[2,0].set_title('Fare vs Age')
```

```
axes[2,0].set_xlabel('Age')
```

```
axes[2,0].set_ylabel('Fare')

axes[2,1].plot()

axes[2,1].set_xlim(0,1)

axes[2,1].set_ylim(0,1)

plt.tight_layout()

plt.show()
```



5.2.2. Histogram of passenger information of

Titanic 01:19

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to black (k).
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'** .
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

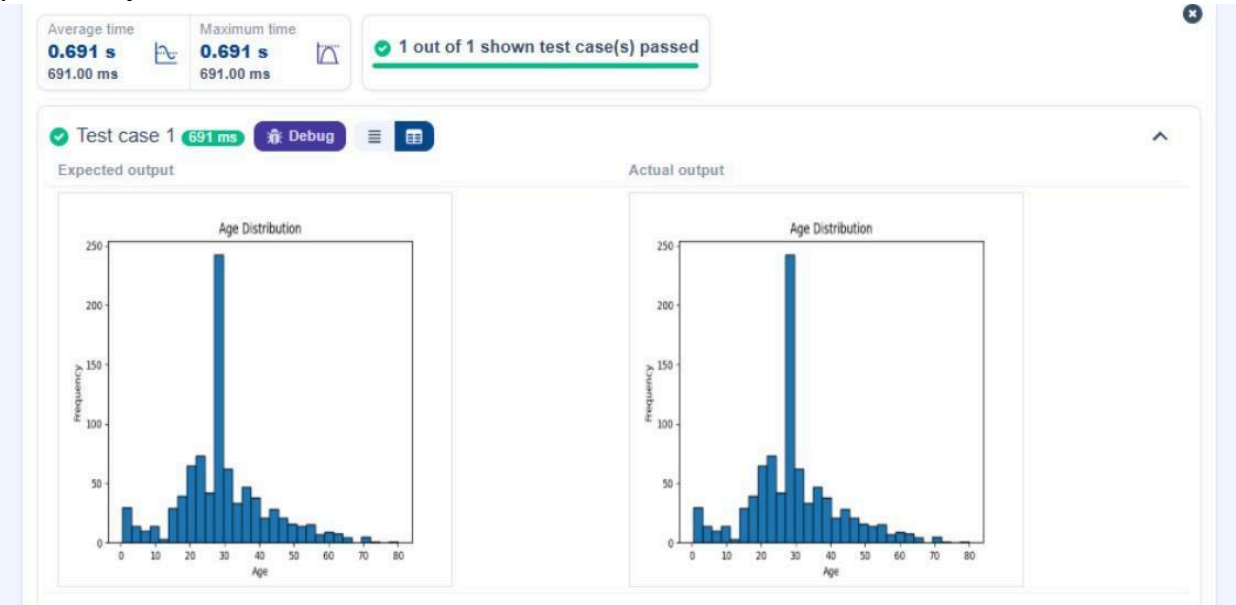
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Write your code here for Histogram
plt.hist(data['Age'], bins=30, edgecolor='k')

plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Age Distribution')
```

plt.show()


Average time
0.691 s
691.00 ms
Maximum time
0.691 s
691.00 ms
1 out of 1 shown test case(s) passed
Test case 1
691 ms
Debug
Expected output
Actual output
Age Distribution
Frequency
Age
Age Distribution
Frequency
Age

5.2.3. Bar plot of survival rate of

passengers 15:39

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived** ' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count** " to the chart.
4. Label the x-axis as '**Survived** ' and the y-axis as '**Count** '.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Write your code here for Bar Plot for Survival Rate
survival_counts=data['Survived'].value_counts().sort_index()
plt.figure()
survival_counts.plot(kind='bar')
plt.title('Survival Count')
plt.xlabel('Survived')
plt.ylabel('Count')
plt.show()
```

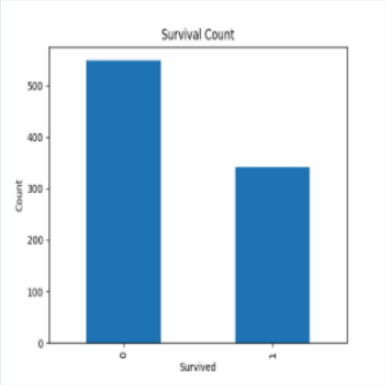
Average time
0.223 s
223.00 ms

Maximum time
0.223 s
223.00 ms

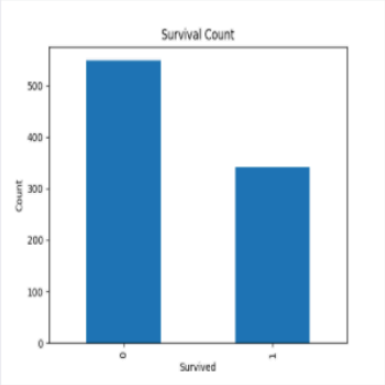
1 out of 1 shown test case(s) passed

Test case 1 223 ms Debug

Expected output



Actual output



5.2.4. Bar Plot for Survival by

Gender 01:21

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **'Sex'** column, then use the **value_ counts ()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Gender"** to the chart.
4. Label the x-axis as **'Gender'** and the y-axis as **'Count'** .
5. The legend should indicate **'Not Survived'** and **'Survived'** .

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

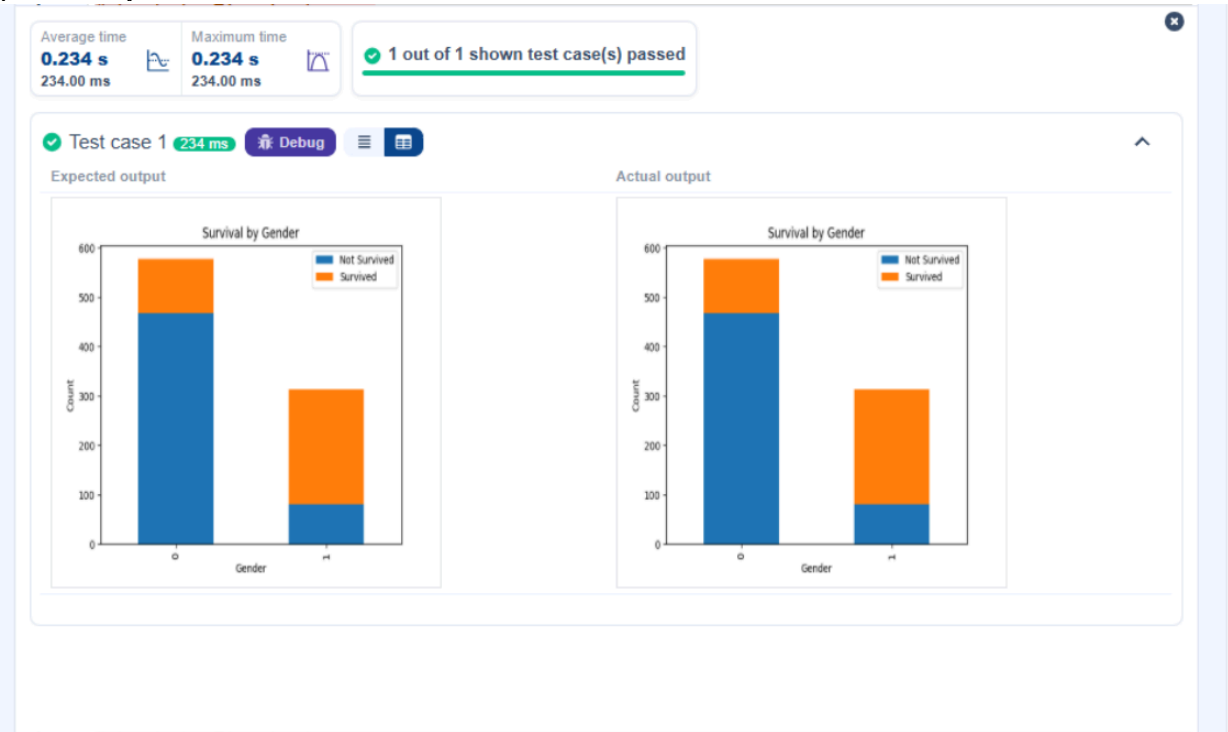
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Write your code here for Bar Plot for Survival by
Gender
counts = data.groupby('Sex')['Survived'].value_counts().unstack()

counts.plot(kind='bar', stacked=True)

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.legend(["Not Survived", "Survived"])
```

plt.show()


Average time
0.234 s
234.00 ms
Maximum time
0.234 s
234.00 ms
1 out of 1 shown test case(s) passed
Test case 1
234 ms
Debug
Expected output
Actual output
Survival by Gender
Not Survived
Survived
Count
Gender
0
1
0
100
200
300
400
500
600

5.2.5. Bar Plot for Survival by

Pclass 01:17

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_ counts ()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title **"Survival by Pclass"** to the chart.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Count'** .
5. The legend should indicate **'Not Survived'** and **'Survived'** .

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note:

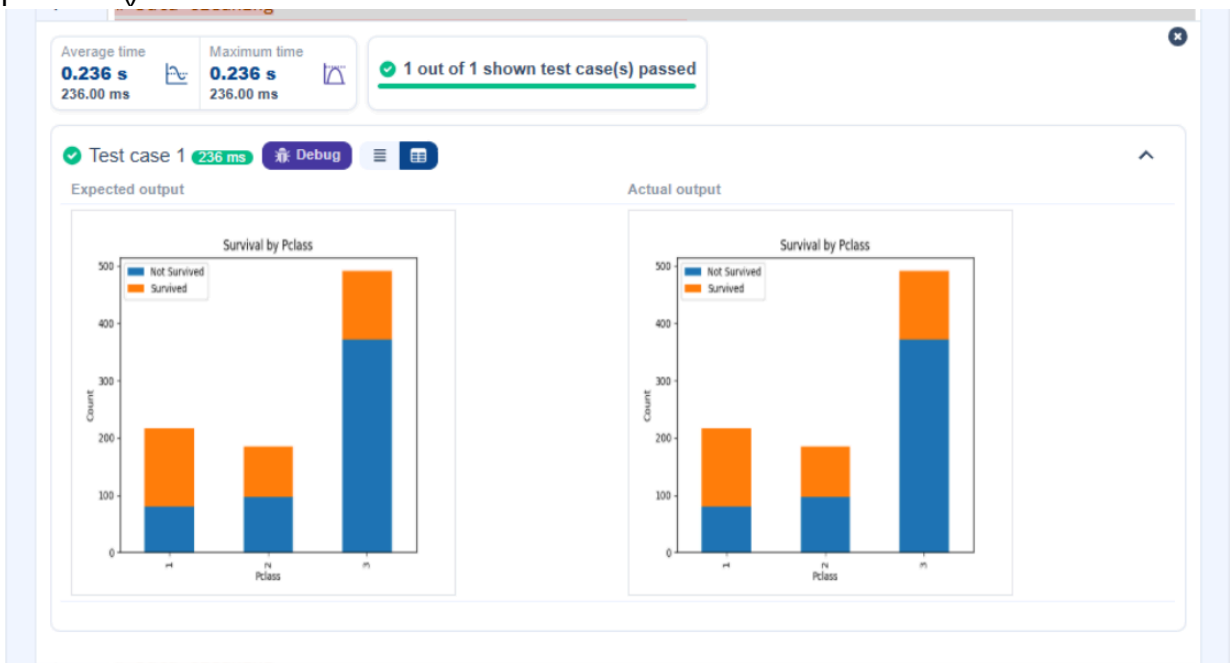
Refer to the visible test case for better reference.

Ensure you use the **groupby ()** function with **value_ counts ()** to count the survivors and non-survivors for each **Pclass** .

Do **not** manually use **size( )** or **unstack( )** without **value_ counts ()**. Use the **value_ counts ()** method for counting survival status directly.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Write your code here for Bar Plot for Survival by
Pclass
counts = data.groupby('Pclass')['Survived'].value_counts().unstack()

counts.plot(kind='bar', stacked=True)

plt.title("Survival by
Pclass") plt.xlabel("Pclass")
plt.ylabel("Count")
plt.legend(["Not Survived", "Survived"])
```

plt.show()


Average time
0.236 s
236.00 ms
Maximum time
0.236 s
236.00 ms
1 out of 1 shown test case(s) passed
Test case 1
236 ms
Debug
Expected output
Actual output
Survival by Pclass
Not Survived
Survived
Count
Pclass
0
100
200
300
400
500
1
2
3

5.2.6. Bar Plot for Survival by

Embarked 04:55

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_ dummies ()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title **"Survival by Embarked "** to the chart.
4. Label the x-axis as **'Embarked'** and the y-axis as **'Count'** .
5. Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

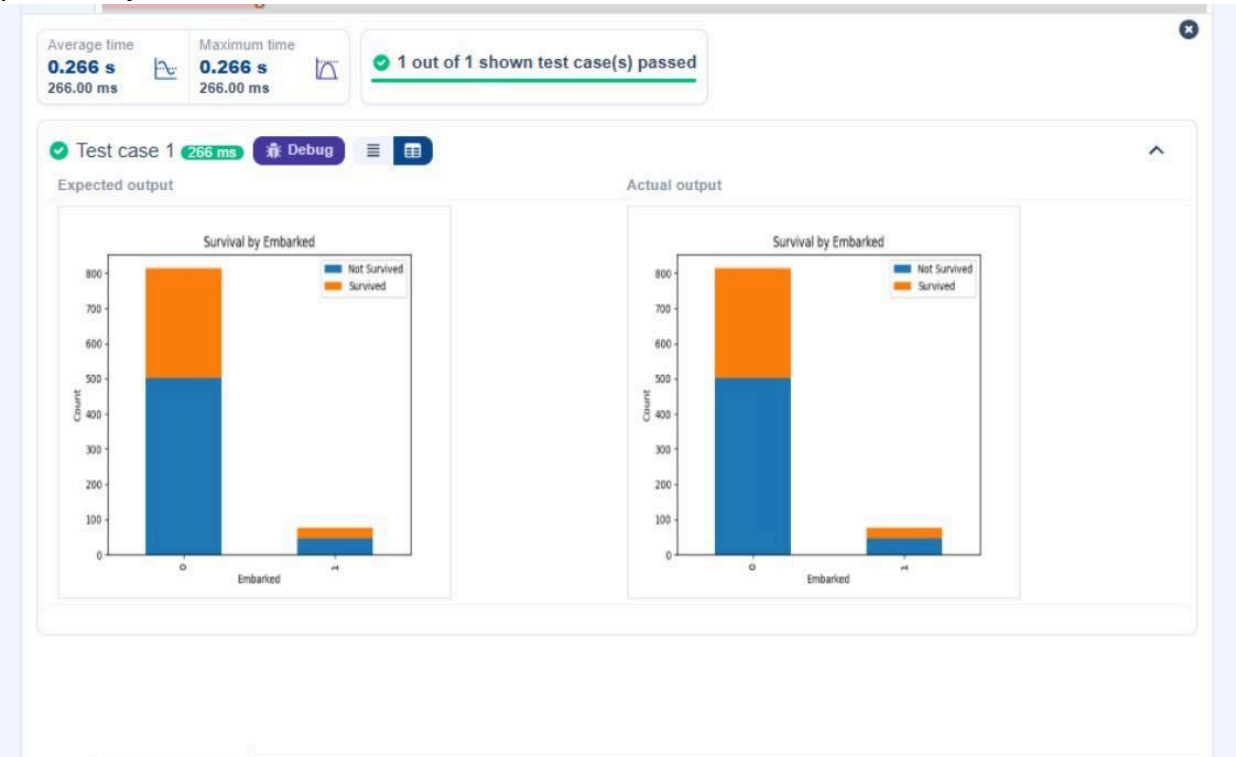
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)



# Write your code here for Bar Plot for Survival by
Embarked grouped =
pd.crosstab(data['Embarked_Q'],data['Survived'])
grouped.plot(kind='bar',stacked=True)
plt.xlabel('Embarked')
plt.ylabel('Count')
plt.title('Survival by Embarked')
plt.legend(['Not Survived','Survived'])
```

plt.show()


Average time
0.266 s
266.00 ms
Maximum time
0.266 s
266.00 ms
1 out of 1 shown test case(s) passed
Test case 1
266 ms
Debug
Expected output
Actual output
Survival by Embarked
Not Survived
Survived
Count
Embarked
0
1
0
100
200
300
400
500
600
700
800

5.2.7. Box plot for Age

Distribution 00:56

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle ('')**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

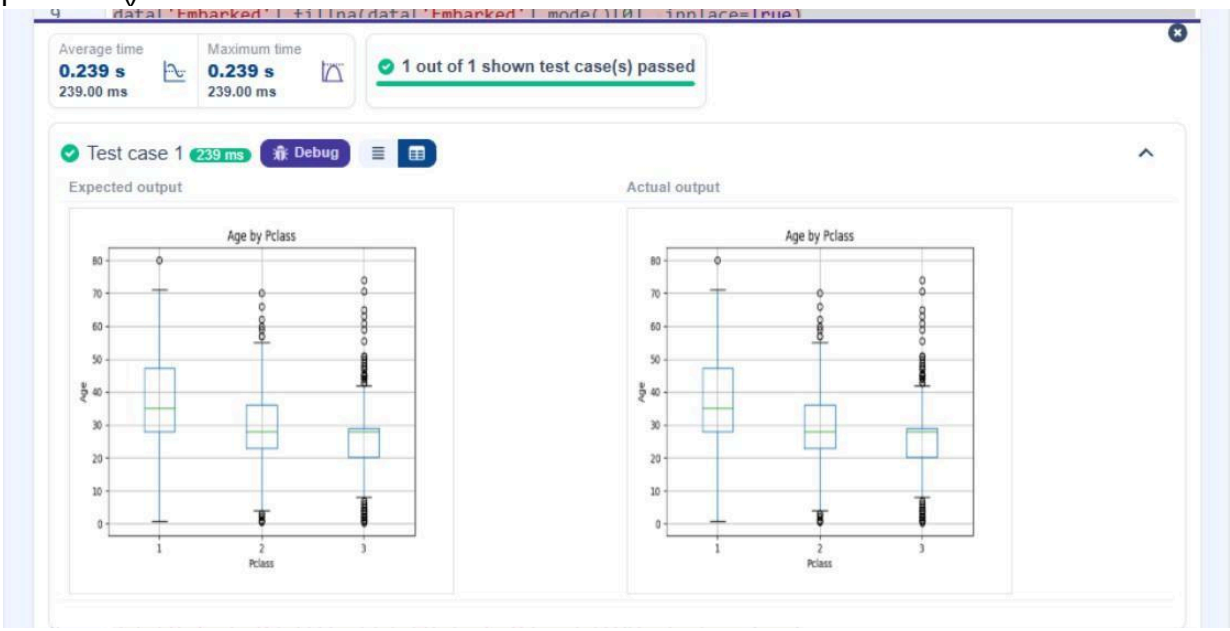
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd

import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Boxplot of Age grouped by Pclass
data.boxplot(column='Age', by='Pclass')

# Remove default subtitle
plt.suptitle('')

# Set title and axis labels
plt.title("Age by Pclass")
plt.xlabel("Pclass")
plt.ylabel("Age")

# Show plot
```

plt.show()


Average time
0.239 s
239.00 ms
Maximum time
0.239 s
239.00 ms
1 out of 1 shown test case(s) passed
Test case 1
239 ms
Debug
Expected output
Actual output
Age by Pclass
Age
Pclass
Age by Pclass
Age
Pclass

5.2.8. Box Plot for Age by

Survived 00:47

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"** .
3. Remove the default subtitle with **plt.suptitle ('')**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

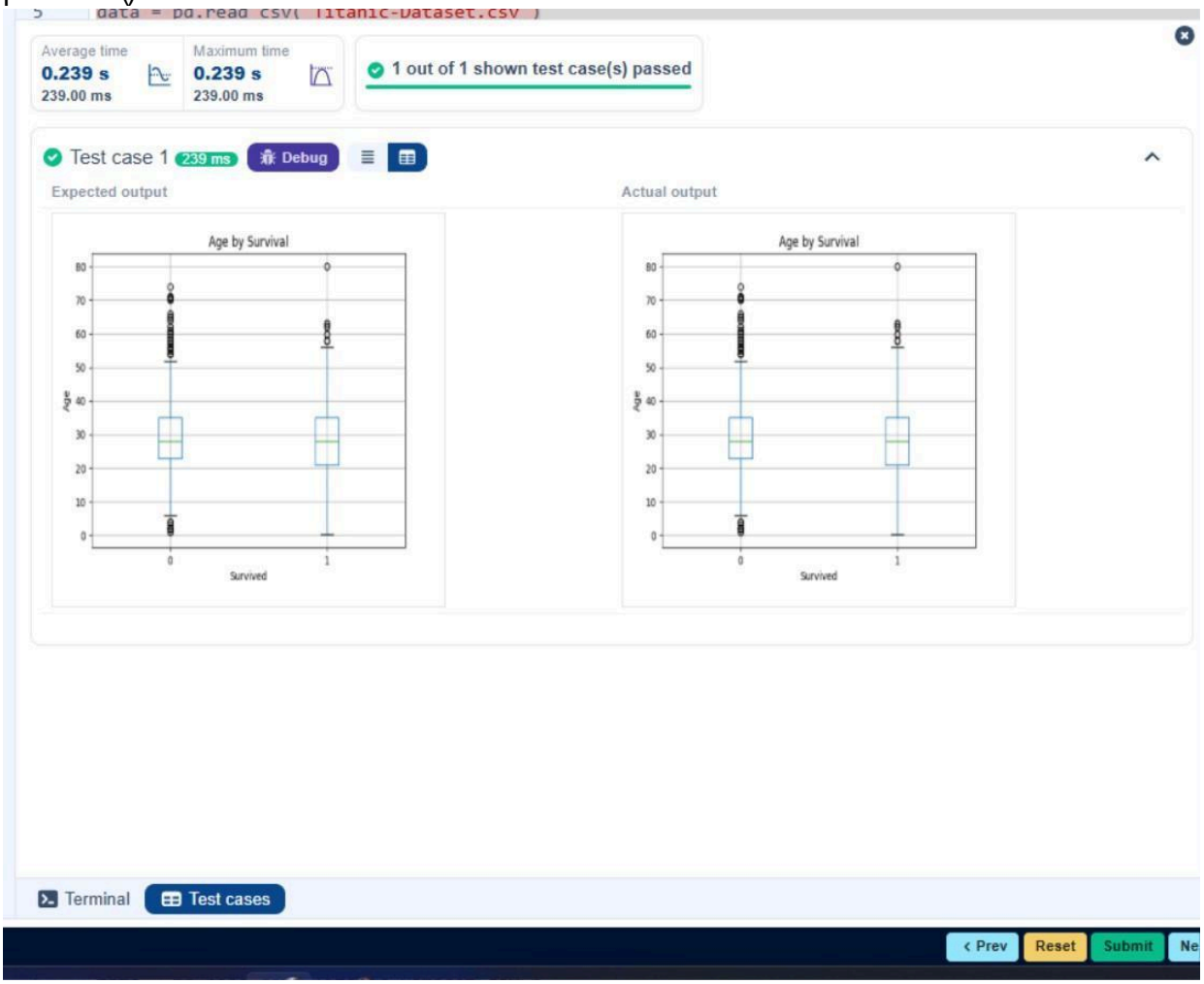
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd

import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Boxplot of Age grouped by Survived
data.boxplot(column='Age', by='Survived')

# Remove default subtitle
plt.suptitle('')

# Set title and axis labels
plt.title("Age by Survival")
plt.xlabel("Survived")
plt.ylabel("Age")

# Show plot
```

plt.show()


data = pd.read_csv('titanic-Dataset.csv')
Average time
0.239 s
239.00 ms
Maximum time
0.239 s
239.00 ms
1 out of 1 shown test case(s) passed
Test case 1
239 ms
Debug
Expected output
Actual output
Age by Survival
Age
Survived
Terminal
Test cases
Prev
Reset
Submit

5.2.9. Box Plot for Fare by

Pclass 01:13

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle (")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

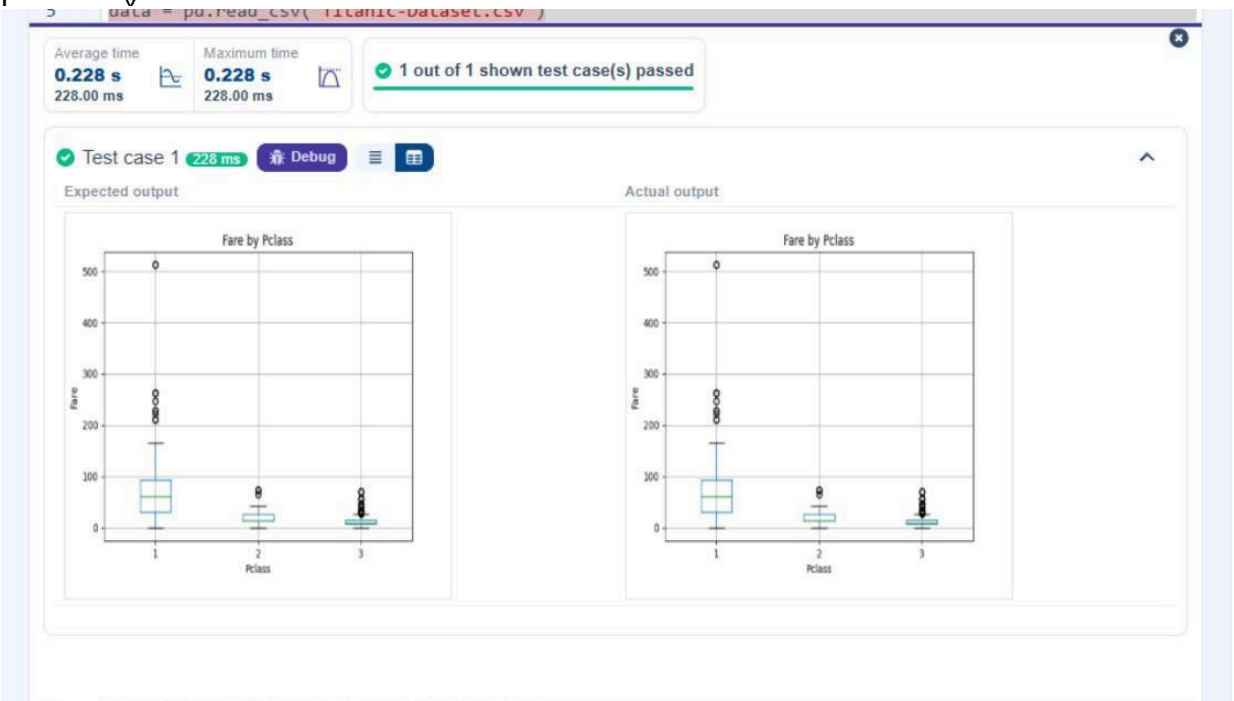
Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # Boxplot of Fare grouped by Pclass
data.boxplot(column='Fare', by='Pclass')

# Remove default subtitle
plt.suptitle('')

# Set title and axis labels
plt.title("Fare by Pclass")
plt.xlabel("Pclass")
plt.ylabel("Fare")

# Show plot
```

plt.show()


Average time
0.228 s
228.00 ms
Maximum time
0.228 s
228.00 ms
1 out of 1 shown test case(s) passed
Test case 1
228 ms
Debug
Expected output
Actual output
Fare by Pclass
Fare
Pclass
Fare by Pclass
Fare
Pclass

5.2.10. Scatter Plot for Age

vs. Fare 05:23

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"** .
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'** .

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) #Write your code here for Box Plot for Fare by
Pclass plt.scatter(data['Age'],data['Fare'])
plt.title('Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
```

plt.show()

Average time
0.173 s
173.00 ms
Maximum time
0.173 s
173.00 ms
1 out of 1 shown test case(s) passed
Test case 1
173 ms
Debug
Expected output
Actual output
Age vs. Fare
Fare
Age
Age vs. Fare
Fare
Age

5.2.11. Scatter Plot for Age vs. Fare by

Survived 28:21

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"** .
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'** .

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female':
1})
data = pd.get_dummies(data, columns=['Embarked'],

drop_first=True) colors = data['Survived'].map({0: 'red',1: 'blue'})

plt.scatter(data['Age'],data['Fare'],c=colors)
plt.title('Age vs. Fare by Survival')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```

```
# Data Cleaning
```

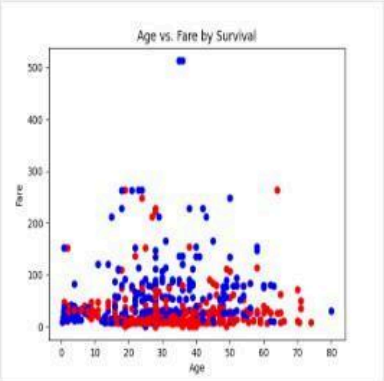
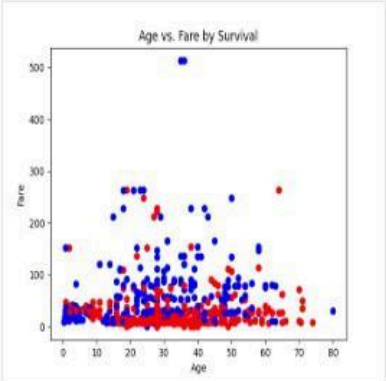
Average time
0.213 s
213.00 ms

Maximum time
0.213 s
213.00 ms

1 out of 1 shown test case(s) passed

Test case 1 213 ms Debug

Expected output



Actual output